Program : **Diploma in Printing Technology**	
Course Code : **5107**	Course Title: **Package Design Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To develop concrete knowledge on the various package design aspects.
- To develop precise hands-on experience and skill in students to design and prepare variety of attractive, efficient and most innovative package models.
- To develop basic knowledge on various packaging software.

Course Prerequisites:

Topic	Course code	Course name	Semester
Drawing Rules, CAD Design		Engineering Graphics	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To do Folding Carton Design.	15	Applying
CO2	To do Design of Flexible & Rigid Packages.	13	Applying
CO3	To effort with Packaging Machines.	5	Applying
CO4	To work with Package design software.	8	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	2			
CO2	3		2	2			
CO3	3						
CO4	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Package Design aspects and Folding Carton Design.**		
M1.01	Familiarization with package terminology and key design aspects.	1	Understanding
M1.02	Prepare the layout and develop- Straight Tuck End (STE) Cartons.	2	Applying
M1.03	Prepare the layout and develop- Reverse Tuck End (RTE) style Cartons.	2	Applying
M1.04	Prepare the layout and develop- Full Seal End (FSE) type Cartons.	2	Applying
M1.05	Prepare the layout and develop- Interlock/Auto-lock bottom type Cartons.	2	Applying
M1.06	Prepare the layout and develop- Tray and Tube style Cartons.	2	Applying
M1.07	Prepare the layout and develop- a corrugated box carton.	1	Applying
M1.08	Design and prepare- Paper Bags/Paper Cup/Envelop.	3	Applying
CO 2	**Design of more Flexible & Rigid Packages.**		
M2.01	Design and Develop common flexible and rigid packages of any kind.	5	Applying
M2.02	Design and Develop innovative and trended flexible/ rigid packages.	4	Applying
M2.03	Design and Develop eco-friendly packages of any kind.	4	Applying
	Lab exam I	2	
CO 3	**To effort with Packaging Machines.**		
M3.01	Study and examine the operation of various packaging machines.	5	Applying

CO 4	**To work with Package design software.**		
M 1.01	Understand the concepts and tools used in 3D-package design software.	4	Understanding
M 1.02	Working with Package Designing Software.	4	Applying
	Lab exam II	2	

Text / Reference:

T/R	Book Title/Author
T1	Brody Aaron L., "The Wiley Encyclopedia of Packaging Technology", John Wiley & Sons, Inc. New York,1997.
R1	Hanlon Joseph F, "Handbook of Package Engineering", CRC Press, USA, 1998.
R2	Chakravarty B, "A Hand Book for Printing and Packaging Technology", Galgotia Publications, 1997.
R4	P I Varghese-Engineering Graphics for Diploma.
R5	Sachidanand Jha- CAD Exercise.

Online Resources:

Sl.No	Website Link
1	https://99designs.com/blog/tips/ultimate-guide-to-product-packaging-design
2	https://landor.com/the-five-fundamentals-of-great-package-design
3	https://design.tutsplus.com/articles/the-beginners-guide-to-packaging-design
4	https://creativemarket.com/blog/the-beginners-guide-to-product-packaging-design
5	https://en.wikipedia.org/wiki/Computer-aided_design